

Tutorial 9-1 Question

Question 1 Bimetallic Strip Expansion

A bimetallic strip of length L is made of two ribbons of different metals bonded together. (a) First assume the strip is originally straight. As the strip is warmed, the metal with the greater average coefficient of expansion expands more than the other, forcing the strip into an arc with the outer radius having a greater circumference (Fig. P18.38). Derive an expression for the angle of bending θ as a function

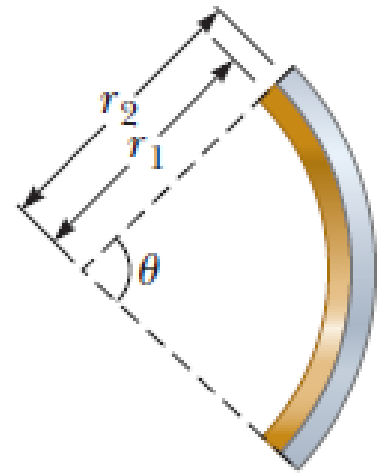


Figure P18.38

of the initial length of the strips, their average coefficients of linear expansion, the change in temperature, and the separation of the centers of the strips ($\Delta r = r_2 - r_1$). (b) Show that the angle of bending decreases to zero when ΔT decreases to zero and also when the two average coefficients of expansion become equal. (c) What If? What happens if the strip is cooled?

Question 2 Coefficient of Volume Expansion

(a) Take the definition of the coefficient of volume expansion to be

$$\beta = \frac{1}{V} \left. \frac{dV}{dT} \right|_{P=\text{constant}} = \frac{1}{V} \frac{\partial V}{\partial T}$$

Use the equation of state for an ideal gas to show that the coefficient of volume expansion for an ideal gas at constant pressure is given by $\beta = 1/T$, where T is the absolute temperature. (b) What value does this expression predict for β at 0°C ? State how this result compares with the experimental values for (c) helium and (d) air in Table 18.1. *Note:* These values are much larger than the coefficients of volume expansion for most liquids and solids.

A vertical cylinder with a heavy piston contains air at 300 K. The initial pressure is $2.00 \times 10^5 \text{ Pa}$, and the initial volume is 0.350 m^3 . Take the molar mass of air as 28.9 g/mol and assume $C_V = \frac{5}{2}R$. (a) Find the specific heat of air at constant volume in units of $\text{J/kg} \cdot ^\circ\text{C}$. (b) Calculate the mass of the air in the cylinder. (c) Suppose the piston is held fixed. Find the energy input required to raise the temperature of the air to 700 K. (d) **What If?** Assume again the conditions of the initial state and assume the heavy piston is free to move. Find the energy input required to raise the temperature to 700 K.

TABLE 18.1 Average Expansion Coefficients for Some Materials Near Room Temperature

Material (Solids)	Average Linear Expansion Coefficient (α)($^\circ\text{C}$) $^{-1}$	Material (Liquids and Gases)	Average Volume Expansion Coefficient (β)($^\circ\text{C}$) $^{-1}$
Aluminum	24×10^{-6}	Acetone	1.5×10^{-4}
Brass and bronze	19×10^{-6}	Alcohol, ethyl	1.12×10^{-4}
Concrete	12×10^{-6}	Benzene	1.24×10^{-4}
Copper	17×10^{-6}	Gasoline	9.6×10^{-4}
Glass (ordinary)	9×10^{-6}	Glycerin	4.85×10^{-4}
Glass (Pyrex)	3.2×10^{-6}	Mercury	1.82×10^{-4}
Invar (Ni-Fe alloy)	0.9×10^{-6}	Turpentine	9.0×10^{-4}
Lead	29×10^{-6}	Air ^a at 0°C	3.67×10^{-3}
Steel	11×10^{-6}	Helium ^a	3.665×10^{-3}

^aGases do not have a specific value for the volume expansion coefficient because the amount of expansion depends on the type of process through which the gas is taken. The values given here assume the gas undergoes an expansion at constant pressure.

Question 3 Heat in a Cylindrical Piston

A vertical cylinder with a heavy piston contains air at 300 K. The initial pressure is 2.00×10^5 Pa, and the initial volume is 0.350 m^3 . Take the molar mass of air as 28.9 g/mol and assume $C_V = \frac{5}{2}R$. (a) Find the specific heat of air at constant volume in units of $\text{J/kg} \cdot ^\circ\text{C}$. (b) Calculate the mass of the air in the cylinder. (c) Suppose the piston is held fixed. Find the energy input required to raise the temperature of the air to 700 K. (d) **What If?** Assume again the conditions of the initial state and assume the heavy piston is free to move. Find the energy input required to raise the temperature to 700 K.

Question 4 Sunspot Radiation

During periods of high activity, the Sun has more sunspots than usual. Sunspots are cooler than the rest of the luminous layer of the Sun's atmosphere (the photosphere). Paradoxically, the total power output of the active Sun is not lower than average but is the same or slightly higher than average. Work out the details of the following crude model of this phenomenon. Consider a patch of the photosphere with an area of $5.10 \times 10^{14} \text{ m}^2$. Its emissivity is 0.965. (a) Find the power it radiates if its temperature is uniformly 5 800 K, corresponding to the quiet Sun. (b) To represent a sunspot, assume 10.0% of the patch area is at 4 800 K and the other 90.0% is at 5 890 K. Find the power output of the patch. (c) State how the answer to part (b) compares with the answer to part (a). (d) Find the average temperature of the patch. Note that this cooler temperature results in a higher power output.

Question 5 Heat due to Kinetic Friction

An iron plate is held against an iron wheel so that a kinetic friction force of 50.0 N acts between the two pieces of metal. The relative speed at which the two surfaces slide over each other is 40.0 m/s. (a) Calculate the rate at which mechanical energy is converted to internal energy. (b) The plate and the wheel each have a mass of 5.00 kg, and each receives 50.0% of the internal energy. If the system is run as described for 10.0 s and each object is then allowed to reach a uniform internal temperature, what is the resultant temperature increase?

Answer

Q1 Derivation & Showing

Q2(b) $3.66 \times 10^{-3} \text{ K}^{-1}$ (c) $3.665 \times 10^{-3} \text{ K}^{-1}$ (d) $3.67 \times 10^{-3} \text{ K}^{-1}$

Q3(a) 0.719 kJ/kg K (b) 0.811 kg (c) 233 kJ (d) 327 kJ

Q4(a) $3.16 \times 10^{22} \text{ W}$ (b) $3.17 \times 10^{22} \text{ W}$ (c) 0.408% (d) $5.78 \times 10^3 \text{ K}$

Q5(a) 2000 W (b) 4.47 °C